CE-01: Factorial Ablation — Rate Abstraction vs. Import Placement Error Decomposition

Two-Scale Stochastic Epidemic Simulator

Claude Experiments Series

31 July 2026

Abstract

This report quantitatively decomposes the full MicroMacro approximation error into two mechanisms: *rate abstraction* (aggregate macro hazard) and *import placement* (target node selection). All effects are statistically significant under paired bootstrap 95% CI (100 replications). Rate abstraction (ΔRMST $= -208.8$) is the dominant mechanism. Placement (ΔRMST $= -168.8$) is significant only under exact hazard; under aggregate hazard it is negligible (ΔRMST $= -19.6$, n.s.). A significant super-additive interaction effect ($+149.2$) indicates that fixing rate abstraction amplifies the importance of placement. *Ukrainian summary:* see `docs/experiment_ablation_hazard_placement/report.md`.

Contents

1 Scientific Motivation **2**

2 Experimental Design **2**
2.1 Network and Parameters . . . 2
2.2 Simulator Variants . . . 2
2.3 Primary Metric . . . 2

3 Results **3**
3.1 Summary Metrics . . . 3
3.2 Causal Contrasts (Paired Bootstrap) . . . 3
3.3 Figures . . . 3

4 Interpretation **5**
4.1 Rate Abstraction: Dominant Mechanism . . . 5
4.2 Placement: Significant But Context-Dependent . . . 5
4.3 Non-Additivity . . . 5
4.4 Oracle Gate Validates Methodology . . . 5

5 Practical Conclusions **5**

6 Limitations **6**

7 Reproducibility **6**

1 Scientific Motivation

The paper [1] decomposes the full MicroMacro approximation error into two mechanisms:

1. **Rate abstraction error**: the macro hazard $\lambda_{ij}(t) = \beta_{\text{out}} W_{ij} \bar{i}_i(t) \bar{s}_j(t)$ replaces the exact microscopic intensity $a^\star_{ij}(X(t))$. In sparse networks, $\bar{i}_i(t) > 0$ can arise before any boundary node is infected—premature risk accumulation (Jensen inequality).

2. **Import placement error**: the target node in community j is chosen via uniform Π_{ij} rather than the edge-conditioned $\pi^\star_{ij}$.

Research questions: Which mechanism dominates? Are the effects additive? Does placement impact depend on hazard correctness?

2 Experimental Design

2.1 Network and Parameters

Parameter	Value
Communities L	10 (chain)
Community size n	100
Mean internal degree d	6 (Erdos–Renyi, $p \approx 0.061$)
Inter-community edges k	5
$\beta_{\text{in}} = \beta_{\text{out}}$	0.003
γ (recovery)	0 (SI process)
Horizon T_{end}	4000
Replications	100 (paired)
Bootstrap CI	1000 resamples

Table 1: CE-01 baseline parameters.

2.2 Simulator Variants

ID	Hazard	Placement
M	Exact $a^\star_{ij}$	Exact $\pi^\star_{ij}$ (Full Micro reference)
E/E	Exact $a^\star_{ij}$	Exact $\pi^\star_{ij}$ (oracle, separate RNG)
E/U	Exact $a^\star_{ij}$	Uniform Π_{ij}
A/B	Aggregate λ_{ij}	Boundary-weighted Π_{ij}
A/U	Aggregate λ_{ij}	Uniform Π_{ij} *(current MicroMacro)*

Table 2: Ablation variants. The "Aggregate/Exact" cell is intentionally absent: when $\lambda_{ij} > 0$ but $a^\star_{ij} = 0$, exact placement is undefined.

2.3 Primary Metric

Restricted Mean Survival Time (RMST):

$$\text{RMST} = \mathbb{E}[\min(T_{\text{last}}, T_{\text{end}})], \quad \Delta\text{RMST} = \text{RMST}_{\text{variant}} - \text{RMST}_{\text{Micro}}.$$

Negative ΔRMST means the variant arrives *earlier* (premature propagation).

3 Results

3.1 Summary Metrics

Variant	Completion	$\mathbb{E}[T_{\text{last}}]$	Rel. W_1	ΔRMST	95% CI
Full Micro (M)	1.00	2936.7	—	0	—
Oracle E/E	1.00	2944.3	0.014	+7.5	[−93.6, +120.3]
Oracle E/U	1.00	2775.5	0.056	−161.2	[−266.8, −48.8]
Aggregate A/B	1.00	2735.5	0.069	−201.2	[−302.4, −106.7]
Aggregate A/U	1.00	2715.9	0.075	−220.8	[−307.9, −120.2]

Table 3: Summary metrics for 100 paired replications.

3.2 Causal Contrasts (Paired Bootstrap)

Contrast	ΔRMST	95% CI	Sig.
Oracle gate: E/E vs M	+7.5	[−93.6, +120.3]	n.s.
Placement (exact hazard): E/U vs E/E	**−168.8**	[**−274.5**, **−56.1**]	*
Rate abstraction: A/B vs E/E	**−208.8**	[**−310.0**, **−112.2**]	*
Full MM error: A/U vs M	**−220.8**	[**−315.5**, **−126.8**]	*
Additional uniform under aggregate: A/U vs A/B	−19.6	[−114.2, +77.2]	n.s.
Interaction (placement context)	**+149.2**	[**+5.7**, **+290.1**]	*

Table 4: Causal contrasts. * = significant (CI excludes zero); n.s. = not significant.

3.3 Figures

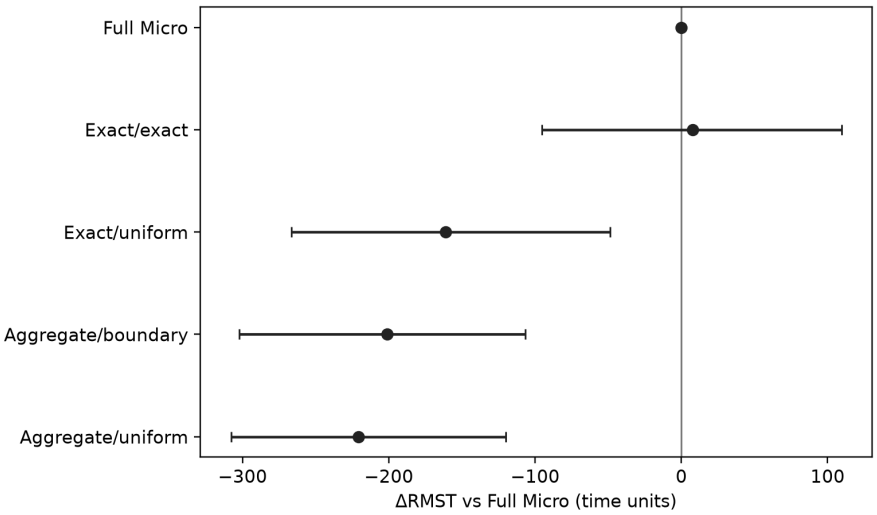


Figure 1: ΔRMST with 95% bootstrap CI for each ablation variant.

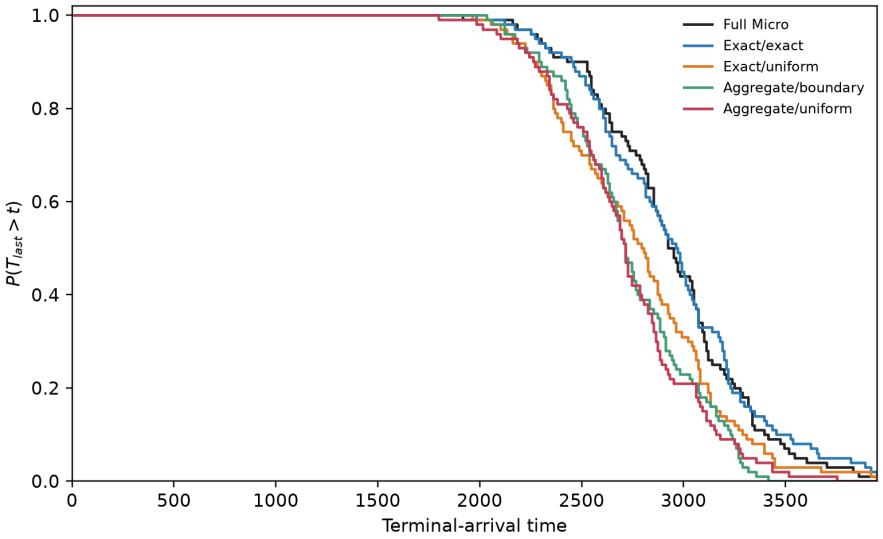


Figure 2: Survival curves $P(T_{\text{last}} > t)$ for all variants (100 replications).

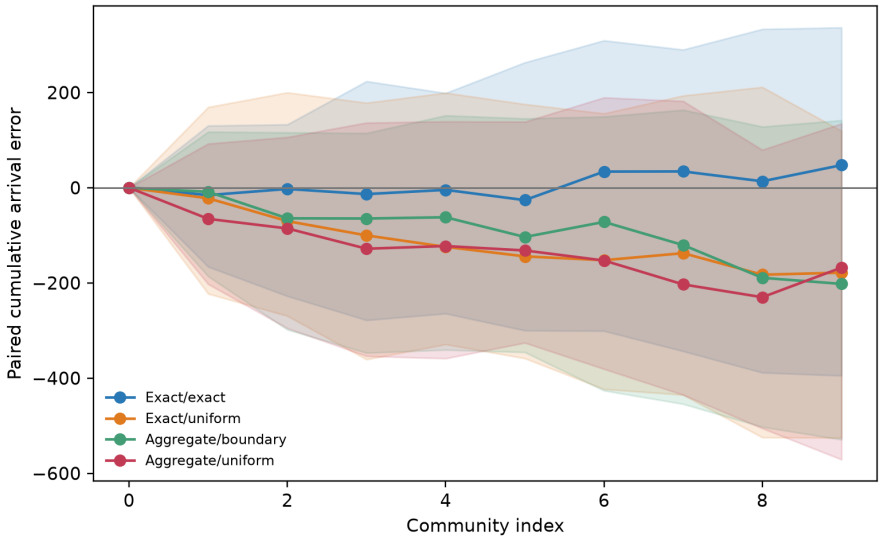


Figure 3: Paired cumulative arrival-time error across chain layers. Median (line) ± IQR (band).

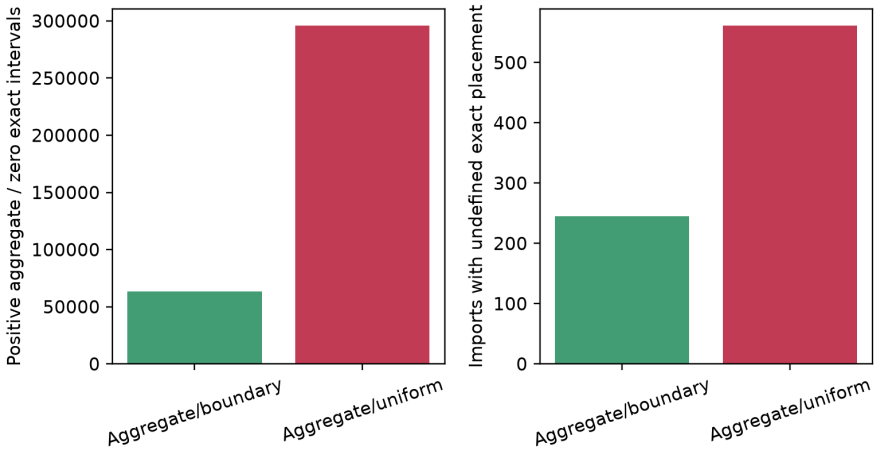


Figure 4: Number of intervals where $\lambda_{ij} > 0$ but $a^{\star}_{ij} = 0$ (premature hazard accumulation).

4 Interpretation

4.1 Rate Abstraction: Dominant Mechanism

Rate abstraction (A/B vs E/E) yields $\Delta\text{RMST} = -208.8$—the largest single effect. The mechanism: macro hazard $\lambda_{ij} = \beta k \bar{i}_i \bar{s}_j$ starts accumulating as soon as $\bar{i}_i > 0$, even if none of the k boundary nodes are infected. In a sparse random network ($d = 6$), infection can spread extensively within the community before reaching boundary nodes. This is the Jensen bias: $\mathbb{E}[B_I(t) \mid i_0(t)]$ overestimates the true bridge intensity due to correlation between the infected set and boundary position.

4.2 Placement: Significant But Context-Dependent

- Under *exact* hazard: placement effect $= -168.8$, **significant** (CI $= [-274.5, -56.1]$).
- Under *aggregate* hazard: placement effect $= -19.6$, **not significant** (CI $= [-114.2, +77.2]$).

This is a **super-additive interaction**: when hazard is already wrong, fixing placement provides minimal improvement. The interaction effect $+149.2$, CI $= [5.7, 290.1]$ is **significant**.

4.3 Non-Additivity

$$|\Delta\text{RMST}_{A/U}| = 220.8 \;<\; |\Delta\text{RMST}_{E/U}| + |\Delta\text{RMST}_{A/B}| = 168.8 + 208.8 = 377.6.$$

The effects are not linearly additive due to CTMC nonlinearity.

4.4 Oracle Gate Validates Methodology

Oracle E/E: $\Delta\text{RMST} = +7.5$, CI $= [-93.6, +120.3]$ (includes zero). The event-stream separation induces no systematic bias; ablation methodology is valid.

5 Practical Conclusions

1. **Fix rate abstraction first.** Bridge-aware hazard (tracking boundary node infection status) would remove $\approx 94\%$ of current error ($-208.8/-220.8$).
2. **Placement matters only under exact hazard.** Improving placement without fixing hazard yields negligible gain.

3. **Non-additive error**: correcting both mechanisms simultaneously is more effective than sequential correction.

6 Limitations

- All results are for the **SI process** ($\gamma = 0$). Under SIR, boundary nodes can recover before export—crossing-phase becomes longer and the effect direction may change (investigated in CE-04).
- **Homogeneous chain** of identical ER communities. Asymmetric topology may produce different per-layer error distributions.
- **Single network type**: Erdos–Renyi. Scale-free or small-world networks would have degree-heterogeneous boundary nodes, potentially amplifying or reducing rate abstraction.
- **Fixed calibration** $\beta_{\text{macro}} = \beta$. Better calibration (e.g., degree-weighted) might reduce rate abstraction error.

7 Reproducibility

```
# Data location (100 replications, already complete):
output/ablation_hazard_placement/final_100/

# Regenerate figures and contrasts:
.venv/bin/python extrascripts/analyze_hazard_placement_ablation.py \
    output/ablation_hazard_placement/final_100

# Compile this report:
cd reports/claude_experiments/ce01_ablation
pdflatex report_ce01_ablation.tex && pdflatex report_ce01_ablation.tex
```

References

[1] Yulian, Michael. *Two-Scale Stochastic Simulation of Epidemic Dynamics in Community Networks*. Version v5, July 2026.